Network Traffic Analysis Report

Credential Transmission Analysis Using Wireshark

1. Objective

The objective of this lab was to capture and analyze network traffic using **Wireshark** in order to determine whether login credentials are transmitted securely or in plain text over a network.

2. Lab Environment

- **Tool Used:** Wireshark
- **Network Interface:** Wi-Fi adapter
- **Target Website:** testfire.net
- **Protocol Analyzed:** HTTP (port 80)
- **Operating System:** Windows 10/11

3. Methodology

Step 1: Packet Capture Initialization

I launched Wireshark and configured it to capture traffic on the **active Wi-Fi interface**. I ensured that packet capture was started before initiating any web activity.

Step 2: Web Interaction

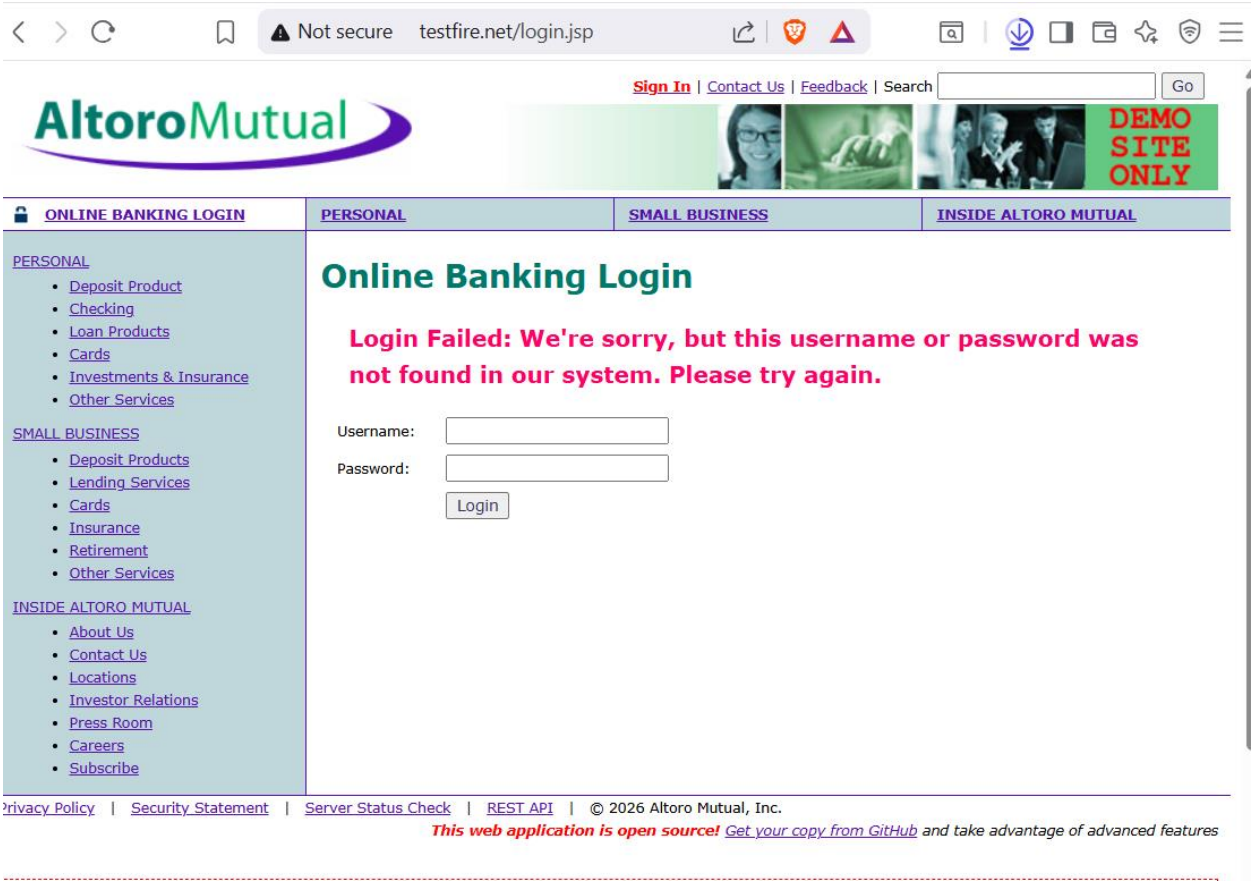
I opened a web browser and navigated to **testfire.net**, where I accessed the login page.

Step 3: Credential Submission

I entered the following test credentials:

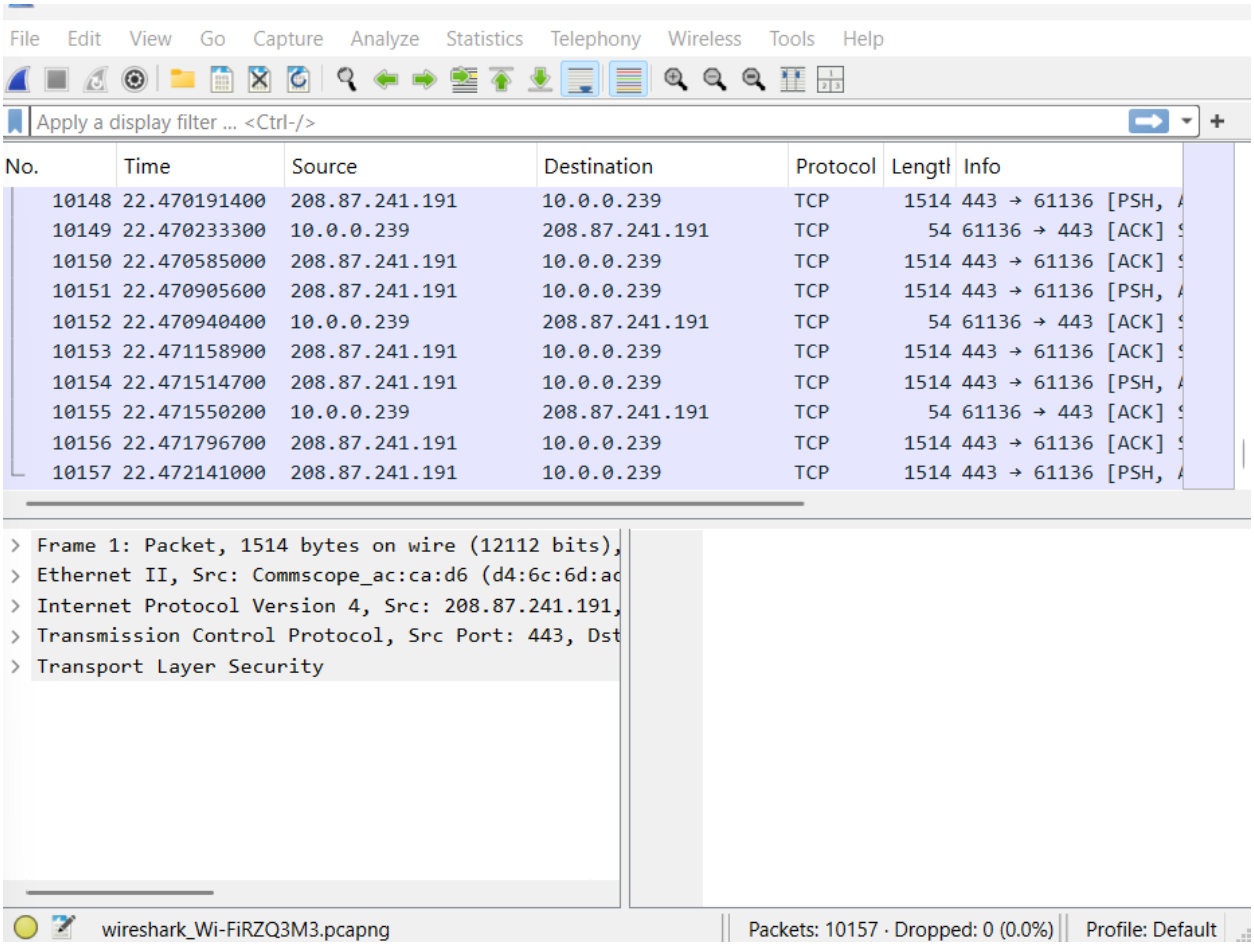
- Username: *AyobamiOdulaja*
- Password: *Ajoke1311@*

I then submitted the login form while the packet capture was ongoing.



Step 4: Capture Termination

After submitting the login request, I stopped the packet capture in Wireshark.



No.	Time	Source	Destination	Protocol	Length	Info
10148	22.470191400	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [PSH, A
10149	22.470233300	10.0.0.239	208.87.241.191	TCP	54	61136 → 443 [ACK] S
10150	22.470585000	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [ACK] S
10151	22.470905600	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [PSH, A
10152	22.470940400	10.0.0.239	208.87.241.191	TCP	54	61136 → 443 [ACK] S
10153	22.471158900	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [ACK] S
10154	22.471514700	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [PSH, A
10155	22.471550200	10.0.0.239	208.87.241.191	TCP	54	61136 → 443 [ACK] S
10156	22.471796700	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [ACK] S
10157	22.472141000	208.87.241.191	10.0.0.239	TCP	1514	443 → 61136 [PSH, A

Step 5: Traffic Filtering & Analysis

I applied the following display filter to isolate relevant traffic:

http

I focused my analysis on HTTP POST requests, particularly those related to login endpoints such as /doLogin.

4. Findings

4.1 Identification of Login Request

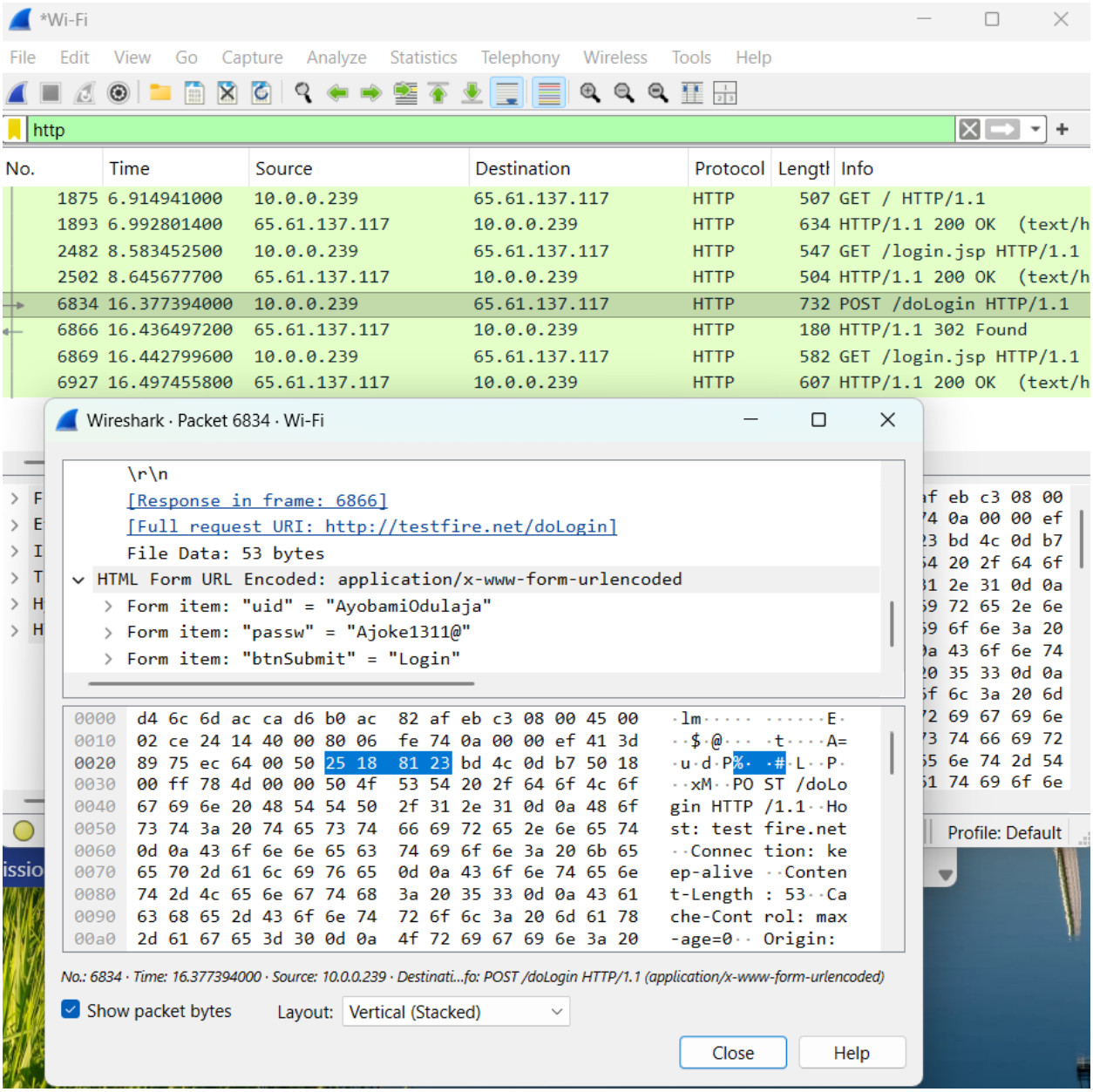
During my analysis, I identified an HTTP **POST request** sent to the endpoint:

/doLogin

4.2 Credential Exposure

Within this request, I observed that the login credentials were transmitted in plain text:

Form item: "uid" = "AyobamiOdulaja"
Form item: "passw" = "Ajoke1311@"



4.3 Data Transmission Format

I also noted that the credentials were transmitted using the following format:

application/x-www-form-urlencoded

This encoding method does not provide encryption, meaning the data remains readable.

5. Security Analysis

From my analysis, I determined that:

- The credentials were transmitted over **HTTP (port 80)**
- No encryption (such as TLS/SSL) was used

This means that the data is exposed in plain text and can be intercepted by malicious actors.

6. Risk Assessment

Risk Factor	Impact
Plain text credential transmission	High
Susceptibility to interception	High
Lack of encryption	Critical

Based on these findings, I assess the overall risk level as **HIGH**.

7. Recommendations

Based on my findings, I recommend the following:

7.1 Use HTTPS

The application should implement HTTPS to ensure all communications are encrypted using TLS.

7.2 Enforce Secure Transmission

All HTTP traffic should be redirected to HTTPS, and insecure endpoints should be disabled.

7.3 Protect Credentials

Passwords should be securely stored using hashing and salting techniques.

7.4 Additional Security Controls

- Implement HSTS (HTTP Strict Transport Security)
- Use secure cookies and proper session management

8. Conclusion

Through this analysis, I confirmed that login credentials were transmitted in plain text over the network. This represents a critical security vulnerability, as sensitive information can be easily intercepted.

This lab reinforced the importance of using secure communication protocols such as HTTPS to protect user data during transmission.